



Graphene-oxide-induced lamellar structures used to fabricate novel composite solid-solid phase change materials for thermal energy storage

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HIGHLIGHTS

- The introduced graphene oxide induced a novel regular lamellar structure for the composite PCMs.
- The heat of melting of the composite PCMs can reach 158.2 J/g.
- The composite PCMs maintains a good thermal stability after 100 thermal cycles.
- Thermal conductivity of the composite PCMs can be improved by 41.4% with 9 wt% graphene oxide.

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ABSTRACT

To improve the thermal conductivity and thermal stability of the phase change materials (PCMs), a series of novel polymeric solid-solid PCMs (SSPCMs) were successfully designed and synthesized with lamellar structures by using poly(ethylene glycol) (PEG) as the phase change ingredient, 4,4'-diphenylmethane diisocyanate (MDI) as the cross-linking agent, and graphene oxide (GO) as the skeleton material. In the preparation, GO nanosheets were introduced into the molecular structure of the SSPCMs through both self-assembly and grafted polymerization to enhance the thermal conductivity and comprehensive thermal properties. Transmission electron microscopy (TEM) and scanning electron microscopy (SEM) images reveal that the SSPCMs possess a novel, regular, lamellar structure consisting of GO sheets with an interconnected network structure, with PEG homogeneously intercalated into the sheets. This molecular structure supplies the SSPCMs with excellent phase change behavior, good thermal cycling stability, and high thermal conductivity. As a result, the SSPCMs display a fast thermal-response rate and outstanding thermal regulating performance, which can maintain their temperature in the range of 50 °C–57 °C for roughly 410 s. Herein, the novel SSPCMs provide a promising potential for applications in thermal management and thermal energy storage systems.

1. Introduction

With the ever-growing energy demand and serious environmental pollution from the burning of fossil fuels, the development of high-performance, low-cost, and environmental-friendly energy storage techniques have become increasingly important [1–3]. Hence, thermal energy storage based on phase change materials (PCMs) has drawn much attention in academia and industry alike [4,5]. PCMs can be used to absorb and release thermal energy over a defined temperature range due to both their high energy-storage capacity and their isothermal

nature during heat storage and release processes. Therefore, PCMs can bridge the gap between energy availability and energy usage, reducing energy waste and improving energy efficiency [6].

Generally, PCMs can be arranged into solid-liquid PCMs and solid-solid PCMs according to their working states [7]. At present, solid-liquid PCMs have been extensively developed for thermal energy storage due to their desirable characteristics: high latent heat capacity, non-reactivity, low or even no phase segregation, supercooling, and good nucleation properties. Nevertheless, special packaging techniques are still necessary to encapsulate the solid-liquid PCMs to prevent their

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fluidity and leakage during phase change processes, which is the challenge in their usage for practical applications [7]. Moreover, packaging techniques with extra storage containers or supporting materials may increase the cost of PCMs and decrease their energy storage density [8]. In contrast, polymeric solid-solid PCMs (SSPCMs) exhibit remarkable properties, such as no liquid generation, no encapsulation requirements, ease of use, and small volume changes due to their solid state during phase transition process between two different solid crystal structures that avoids the leakage problem. Furthermore, they exhibit excellent design flexibility and long lifespans, making them able to be used conveniently and with various shapes and dimensions. Consequently, SSPCMs, an extremely attractive technology, have been well-developed by chemical grafting, blocking, or crosslinking copolymerization [9–12]. Usually, solid-liquid PCMs with desirable properties and high reactivities are selected as both the energy storage functional component and soft segment of SSPCMs, while polymer skeletons or hard segments are introduced to the molecular chains of the SSPCMs as supporting processes to keep the material in solid state during phase change processes [13,14].

In particular, poly(ethylene glycol) (PEG), a linear polymer and a promising solid-liquid PCM, has attracted considerable attention for its high latent heat storage capacity, diversified molecular weight, excellent biocompatibility, non-toxicity, and good thermal/chemical stability. Thus, PEG and its derivatives are extensively utilized as the phase change ingredient while designing polymeric SSPCMs due to their outstanding properties and the high reactivity of its terminal hydroxyl groups. Recently, various SSPCMs based on PEG have been prepared for thermal energy storage applications, such as polystyrene-graft-PEG copolymers [9], melamine/formaldehyde/PEG cross-linking copolymers [10], linear polyurethane ionomers [11], β -cyclodextrin/4,4'-diphenylmethane diisocyanate (MDI)/PEG cross-linking copolymers [12], and polyurethane-urea copolymers [13]. Unfortunately, the heat of phase transition of SSPCMs still needs to be significantly improved for thermal energy storage. Current synthesis methods involve the PEG chains of SSPCMs almost always being linked with their hard segments by functional groups, hindering the mobility of PEG chains, and causing the arrangement and orientation of PEG chains and resulting in a decrease in the heat storage capacity of SSPCMs. Consequently, it is necessary to reduce the influence of the hard segments on PEG and improve the latent heat of SSPCMs. Moreover, SSPCMs are still limited by low thermal conductivity, which seriously decreases the thermal energy-storage efficiency and hinders their practical applications. Hence, it is also essential to enhance the thermal conductivity of novel SSPCMs.

Therefore, to enhance the thermal conductivity, heat capacity during phase transition, and thermal stability of these materials, this study explores a novel SSPCM with a three-dimensional lamellar structure, fabricated by introducing graphene oxide (GO) into polymer molecular chains through grafted polymerization and self-assembly. Recently, graphene has garnered much attention for its high specific surface area ($\geq 2600 \text{ m}^2/\text{g}$) and superior thermal conductivity of $5000 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$ [15]. Thus, graphene has been used widely as a filler to fabricate highly thermally conductive polymer composites, such as graphene/polyethylene composites [16], graphene/polystyrene composites [17] and graphene/poly(vinyl alcohol) [18]. In addition, graphene is light in weight and possesses inherently good flexibility, a high Young's modulus (1100 GPa), and a large ultimate strength (116 GPa). Consequently, graphene has broad applications due to these excellent properties. Therefore, in this study, graphene was considered as an ideal material to improve the thermal conductivity of SSPCMs. It is well-known that GO has several oxygen-containing functional groups on its surface, such as epoxy, carbonyl, and hydroxyl groups [19]. These functional groups provide compatibility between GO and polymers [20]. Thus, a novel design concept for cross-linked polymeric SSPCMs with good comprehensive thermal performances was proposed, where PEG was utilized as the soft chain and phase change functional units, MDI was chosen as a blocking agent, and GO was selected as a hard-

molecular skeleton. This procedure formed a three-dimensional, interconnected framework in polymeric SSPCMs through both hydrogen bonding and electrostatic interactions between functional groups. As a result, a regular lamellar structure of GO was introduced into the polymer molecular chains. This created the thermally conductive pathways for the interior PEG chains and ensured the complete storage and release of thermal energy by PEG during phase change procedures, resulting in a high thermal capacity and good thermal stability for the novel SSPCMs. Additionally, this three-dimensional, interconnected structure can hold PEG without leakage during phase change processes. Finally, GO also provided good mechanical properties for the novel SSPCMs.

As a result, this current work, presents a novel design concept for cross-linking polymeric SSPCMs with good comprehensive thermal performances. The novel SSPCMs, with regular lamellar structures, were synthesized through a graft-crossing method by using PEG as a phase change ingredient. MDI was selected as a hard-molecular chain and GO was employed as the supporting skeleton for PEG. The introduction of GO nanosheets may induce a lamellar structure for the SSPCMs through self-assembly during polymerization. Herein, the fabrication mechanism, microstructure, and crystalline behaviors of the polymeric SSPCMs were investigated in detail. Furthermore, the influences of both the lamellar structure and the introduction of the GO on the thermal energy storage property, thermal stability, and the thermal conductivity of the SSPCMs were explored. The concept illustrated in this study hopes to develop prospective composite PCMs for applications in thermal energy storage systems.

2. Materials and methods

2.1. Materials

PEG (with an average molecular weight of 8000 g/mol), potassium permanganate (KMnO_4), and MDI were obtained from Tianjin Alfa Aesar Co. (Tianjin, China). Flake graphite with an average particle size of 325 mesh was purchased from XFNANO Materials Tech Co. Ltd. (Nanjing, China). Concentrated sulfuric acid (H_2SO_4) with a purity of 98%, hydrogen peroxide (H_2O_2) with a purity of 30%, sodium nitrate (NaNO_3), and anhydrous N, N-dimethylformamide (DMF) were commercially supplied by Xilong Chemical Co. Ltd. (Shantou, China).

2.2. Fabrication of PEG/MDI/GO SSPCMs

First, GO was prepared from flake graphite using a modified Hummers' method [19,21]. The synthesis of novel SSPCMs was conducted through a two-step polymerization process. First, PEG and DMF were mixed at 85°C and stirred in a three-neck flask under an argon atmosphere. Then, a predetermined amount of MDI (PEG:MDI = 1:2, molar ratio) was dissolved in DMF to produce a homogeneous solution that was slowly added into the PEG solution. Next, the mixture reacted at 85°C for 5 h to obtain a polyurethane prepolymer. Then, GO was dissolved in DMF and slowly dropped into the prepolymer solution. The reaction was allowed to continue for 24 h to accomplish self-assembly and grafted polymerization. Finally, the mixture was dried to remove the DMF, obtaining the final product. The added amounts of GO were 3, 6, 9, and 12 wt% compared with the amount of PEG, with the obtained composites referred to SSPCM-1, SSPCM-2, SSPCM-3 and SSPCM-4, respectively.

2.3. Characterization of the obtained PEG/MDI/GO SSPCMs

Scanning electron microscopy (SEM) images of the prepared SSPCMs were taken on a SEM (JSM-6360LV, JEOL Ltd., Japan) operating at an accelerating voltage of 20 KV. The samples were sputtered with a gold-palladium alloy before testing. The microstructures of the SSPCMs were detected by transmission electron microscopy (TEM)

(Hitachi JEM-1200EX, JEOL Ltd., Japan) operating at an acceleration voltage of 120 kV. The SSPCM samples were dispersed in ethanol, with some pieces collected on carbon-coated 300-mesh copper grids for TEM observation. The Fourier transform-infrared spectroscopy (FT-IR) spectra were obtained using a Nicolet 6700 spectrometer (USA) with KBr pellets, in the wavenumber range of 4000–400 cm^{-1} . The X-ray diffraction (XRD) patterns of the powders were obtained using an automated X-ray powder diffractometer (XRD, D8 Advance, Bruker, Germany) with monochromated $\text{Cu-K}\alpha$ radiation ($\lambda = 1.5418 \text{ \AA}$). The XRD patterns were obtained at a scanning rate of $0.02^\circ/\text{s}$ between 2θ of 5° and 80° . Raman spectra of the samples were obtained with a LabRAM HR Evolution spectrometer (Horiba JY, France) using a laser excitation of 532 nm. An X-ray photoemission spectrometer (XPS, Thermo ESCALAB 250Xi) with an $\text{Al K}\alpha$ ($h\nu = 1486.6 \text{ eV}$) X-ray source was used to identify the chemical composition and surface functionalities of the SSPCMs. The melting and crystalline behaviors of the SSPCMs were measured by a differential scanning calorimeter (DSC, Setaram Sensys, France), operating at a heating and cooling rate of $5^\circ\text{C}/\text{min}$ under a nitrogen atmosphere. The samples were first heated to 120°C and held at this temperature for 5 min to eliminate the influences of heat and processing history before testing. Thermal cycling stability experiments of the samples were carried out consecutively up to 100 thermal cycles using the DSC to determine the thermal reliability of the SSPCMs in terms of the change in phase change temperatures and latent heat values. Furthermore, temperature-time curves of pure PEG and SSPCM-3 were investigated as described in our previous work [21]. The measurements were carried out from -10°C to 100°C at a heating and cooling rate of $5^\circ\text{C}/\text{min}$. *In-situ* XRD was performed to determine the crystalline properties of the synthesized SSPCMs in the same temperature range as that of the DSC measurements. Diffraction patterns of the samples were recorded at a series of temperature intervals during the heating process. Before each measurement, the sample was equilibrated for roughly 3–5 min. Thermal conductivity measurements were performed using a thermal conductivity analyzer (Setaram, TCI) at room temperature. Reported results demonstrate the average of three measurements for each sample to ensure repeatability. To evaluate the thermal response rate of the SSPCMs, photographs were taken by a thermographic camera (Fluke Ti32, USA) to record the temperature distribution of the samples, which were placed on a hot stage at a constant temperature of 60°C .

3. Results and discussion

3.1. Design and synthetic mechanism of the SSPCMs

A schematic of the mechanism to prepare the novel SSPCMs is represented in Fig. 1. First, PEG and MDI with a predetermined molar ratio (1:2) were separately mixed in DMF. Then, the MDI solution was added into the PEG solution. The mixed solution was stirred at 85°C for 2 h to form the PEG-MDI prepolymer. Theoretically, there are 2O–H bonds in the PEG, that can react with the isocyanate groups of MDI, resulting in the PEG-MDI prepolymer containing isocyanate groups at the end of its molecular chains. Consequently, different amounts of GO were introduced into the prepolymer solution. As it is well known, there are several oxygen-containing functional groups in GO, such as epoxy, carbonyl, and hydroxyl groups on its surface that can react with the prepolymer and provide compatibility between GO and the PEG-MDI prepolymer. As a result, GO was successfully introduced into the molecular chains of the polyurethane polymer when the reaction went to completion.

Furthermore, the reaction between the functional groups of GO and the PEG-MDI prepolymer can lead to the formation of amides or carbamate esters, as seen from Fig. 1(b). As is known, the carboxyl and hydroxyl functional groups are in the edge and surface of GO, respectively. Consequently, the PEG-MDI prepolymer can be effectively grafted on the surface of GO through self-assembly between the

functional groups. This resulted in the GO nanosheets and the PEG-MDI prepolymer alternately arranged with a lamellar structure. Thus, PEG was homogeneously inserted into the GO nanosheets, with the GO nanosheet being formed into the hard segment of the polymer molecular chain, creating the main skeleton of the polymer. Interestingly, the PEG segments can move within the spaces of GO nanosheets, providing both a soft segment and the presence of functional groups in the polymer. Consequently, GO sheets form a topologically connecting and thermally conductive network within the PEG matrix, providing a high thermal conductivity to the polymers due to the intrinsically high aspect ratio, high specific surface area, and high thermal conductivity of GO. Additionally, the lamellar structure of GO can limit the leakage of the PEG segment during phase change procedures. Therefore, the novel SSPCMs were successfully synthesized through self-assembly via polymerization, with PEG being distributed homogeneously in the three-dimensional, interconnected network of GO nanosheets in the composites. Moreover, the introduction of GO in the composites may induce a high thermal conductivity and good mechanical properties for the SSPCMs due to its inherent chemical nature.

3.2. Micro-morphologies of the SSPCMs

SEM images are used to visually evaluate the morphologies of GO and the SSPCMs. Fig. 2(a) shows the morphology of a representative GO sample obtained in this work. It can be observed that GO sheets present a wrinkled surface with curling edges, similar to previous reported [19,20]. It is worth noting that the structure of the GO sheets plays a beneficial role in the self-assembly reaction. Fig. 2(b) presents the SEM image of pure PEG. In contrast to GO, it is clear that the surface of pure PEG is relatively smooth and uniform. The SEM images of SSPCM-1, SSPCM-2, SSPCM-3, and SSPCM-4 are displayed in Fig. 2(c)–(f). It is significant that PEG was incorporated into the three-dimensional interconnected network of GO with no evidence of serious interfacial separation or cracking being observed, which indicates that the compatibility between PEG and GO is quite good, producing a strong combination between the PEG-MDI prepolymer and GO. Therefore, PEG can be kept in the composites without leakage. Fig. 2(g) is the enlarged view of SSPCM-3. It clearly seen that GO sheets are stacked layer-by-layer, leading to a regular lamellar structure that can be described as a three-dimensional, interconnected network of the composites. The thickness of the GO sheets is increased due to the complete covering of PEG [22]. As a result, GO forms a topologically connecting and thermally conductive network for the interior PEG. This interconnected structure not only ensures that PEG maintains its phase change behavior in the composites but also supplies the PEG with high thermal conductivity. This phenomenon implies that the novel SSPCMs possess a high thermal conductivity due to the three-dimensional, interconnected network structure of the composites.

High-resolution TEM images were also used to investigate the microstructure of the as-formed GO nanosheets and the composites. It is clearly seen from Fig. 3(a) that the transparent GO nanosheets have irregular corrugation and folding in the sheets suggesting that there could be strong interface connections between GO nanosheets and PEG. As shown in Fig. 3(b), the GO nanosheet clearly illustrates an amorphous nature. From Fig. 3(c) and (d), it is significant that the PEG was homogeneously dispersed in the GO nanosheet, causing the formation of the composites' lamellar structure. These results imply that the as-proposed novel SSPCMs were successfully synthesized.

3.3. Chemical structure of SSPCMs

Raman spectroscopy, a powerful tool to investigate structure of the composites, was used to probe the pure PEG, GO, and SSPCMs obtained in this study. As seen from Fig. 4, two representative bands at $\sim 1340 \text{ cm}^{-1}$ (D band) and $\sim 1580 \text{ cm}^{-1}$ (G band) are observed for the GO sample. As is well known, the D to G band peak intensity ratio, I_D/I_G

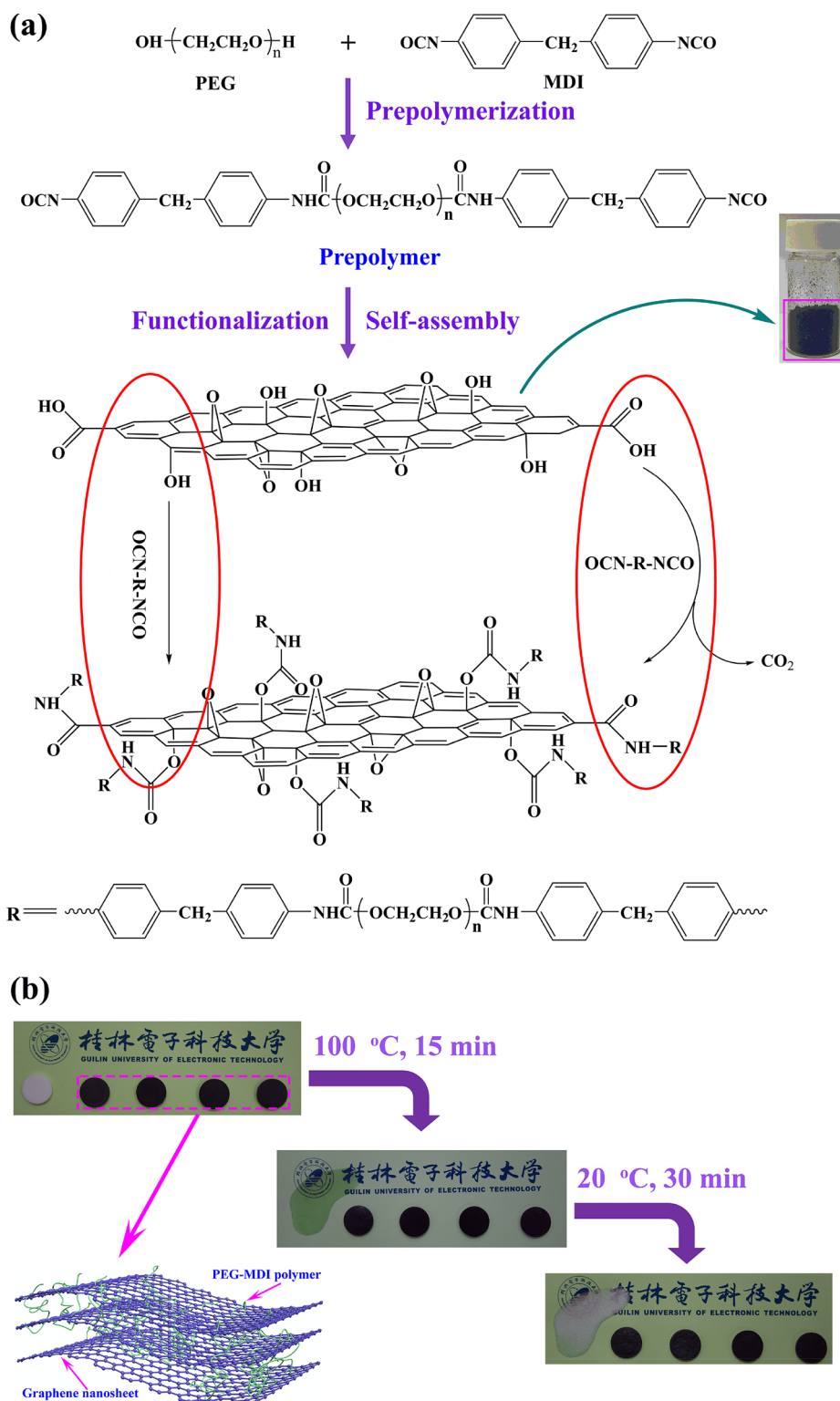


Fig. 1. Schematic illustration for (a) the polyurethane prepolymer and (b) grafted-polymerization and self-assembly process.

I_G , is generally used to verify the degree of defects of carbon nanomaterials [23]. An I_D/I_G ratio of 0.98 is presented from the spectroscopy, quite near 1.0, confirming a regular, defect-free carbon structure of GO. The I_D/I_G for GO decreases from 0.98 to 0.96 in SSPCM-3, indicating the restoration of sp^2 carbon domains and an increase of the graphitic nature of the GO nanosheets in the SSPCMs [24]. In addition, it is significant that the characteristic peaks of PEG continuously disappear with the increase of the GO content, further confirming that PEG was

distributed into the lamellar structure of GO nanosheets, limiting the mobility of the PEG molecular chain so as to hold it without leakage.

Fig. 5 displays the FTIR spectra of GO, PEG, and all SSPCMs. As can be seen from Fig. 5(a), the band appearing at 3416 cm^{-1} is associated with O-H stretching vibrations. The stretching vibration of C=O in carbonyl or carboxyl groups appears at 1722 cm^{-1} . The intense band at 1115 cm^{-1} corresponds to the C-O stretching vibration. These results confirm that hydroxyl, epoxy, and carboxyl groups were introduced

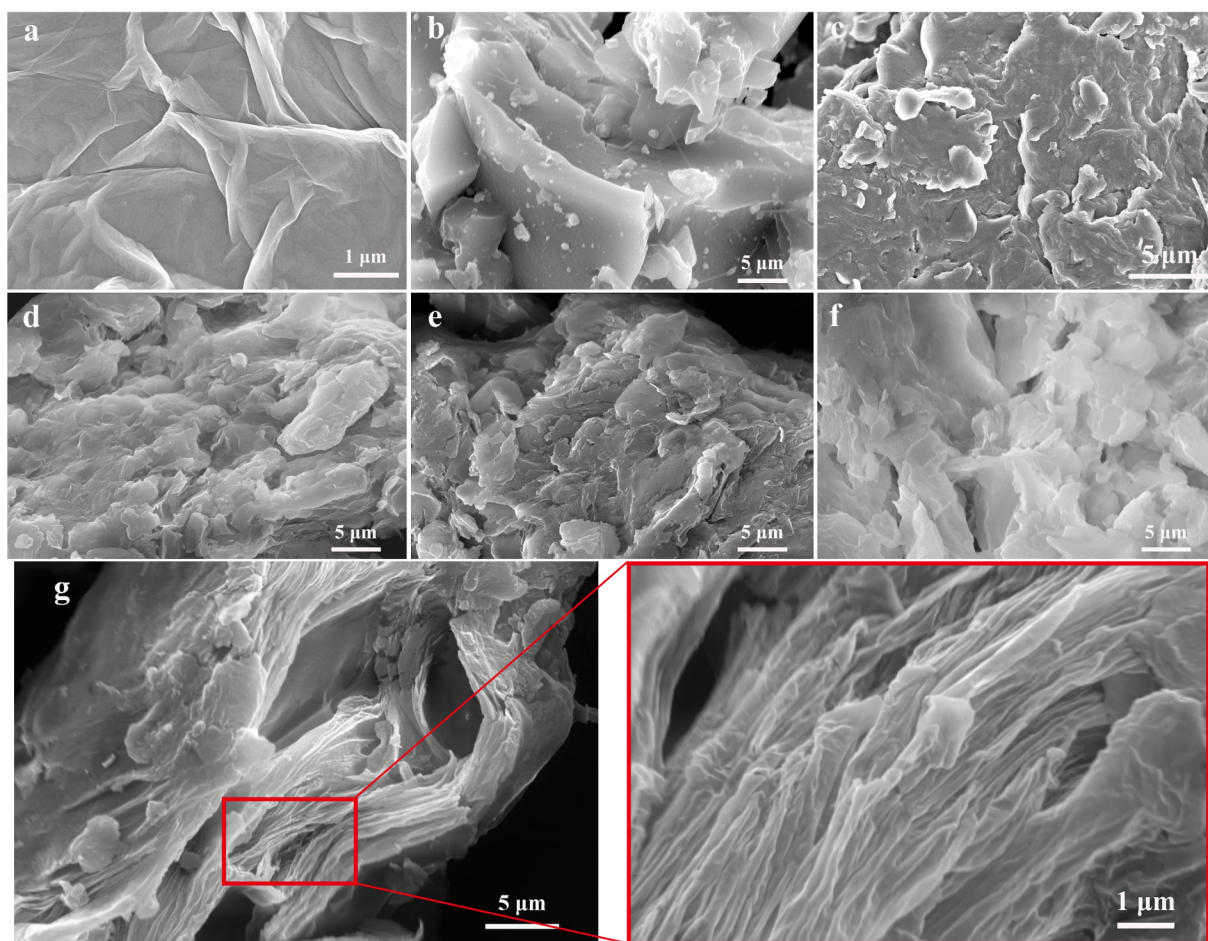


Fig. 2. SEM images of (a) the synthesized GO, (b) pure PEG, the surfaces of (c) SSPCM-1, (d) SSPCM-2, (e) SSPCM-3, (f) SSPCM-4 and (g) the enlarged view of SSPCM-3.

onto the surface of GO via an oxidation process [23]. In the spectrum of pure PEG, the broad band at 3470 cm^{-1} is attributed to the stretching vibrations of O–H groups, while the absorption peaks at 2886 cm^{-1} , 1466 cm^{-1} , 1354 cm^{-1} , 1275 cm^{-1} , 963 cm^{-1} , and 847 cm^{-1} belong to the C–H bonds. The peaks at 1242 cm^{-1} and 1100 cm^{-1} are ascribed to the stretching vibrations of C–O. Compared with the spectra of GO and PEG, it is obvious that the synthesized SSPCMs possess a C–O stretching vibration peak, C–H stretching/bending vibration peaks, and O–H stretching vibrations, which are nearly the same with the characteristic peaks of PEG and GO. Additionally, it is worth noting that the new stretch in the spectrum of SSPCMs appearing at 1645 cm^{-1} can be assigned to an amide carbonyl-stretching mode, while the peak at 1563 cm^{-1} can originate from either amides or carbamate esters corresponding to the coupling of the C–N stretching vibration with the CHN deformation vibration, and also the absorption peak at around of 1700 cm^{-1} is the characteristic absorption peak for the stretching vibration of C=O. Furthermore, the characteristic absorption peak of isocyanate ($2275\text{--}2263\text{ cm}^{-1}$) from MDI disappears completely in the SSPCMs spectra. These characteristic peaks imply the successful reaction between PEG and MDI, which also indicates that GO reacted successfully with phenyl isocyanate and resulted in the formation of hard segments in the molecular chains of the composites. The spectra in Fig. 5(c)–(f) are nearly the same, confirming that the addition of GO has no influence on the chemical structure of the SSPCMs.

XPS was performed to investigate the elemental composition and chemical states of the composites by using SSPCM-3 as a representative sample. It is significant that C, N, and O elements are present in the XPS survey spectrum in Fig. 6(a). The C 1s high-resolution XPS spectrum can

be deconvoluted into three peak components shown in Fig. 6(b), which are associated with non-oxygenated ring carbon (C–C or C=C), epoxy and alkoxy carbon (C–O), and carbonyl carbon (C=O) functional groups. For the O 1s XPS spectrum shown in Fig. 6(c), the strong peak located at 533 eV can be attributed to C=O, while the small peak at 531.9 eV corresponds to C–O. Moreover, the presence of the C–N bond is confirmed by the peak (400 eV) appearing in the N 1s XPS spectrum, as shown in Fig. 6(d). These observations indicate that MDI, PEG, and GO were successfully bonded together by grafted polymerization to form a novel, multi-functional copolymer. Based on these FTIR and XPS results, it can be concluded that the novel SSPCMs were successfully prepared as expected.

3.4. Thermal properties of the SSPCMs

The thermal properties of the synthesized SSPCMs with different amounts of GO were evaluated by DSC analysis. The obtained curves are shown in Fig. 7 and the corresponding data is summarized in Table 1. It is clear that there are strong and sharp melting/crystallization peaks appearing in the DSC curves of PEG observed in Fig. 7(a) and (b), which are due to the excellent solid-liquid phase change properties of PEG. Similarly, there is only one single and well-resolved melting peak appearing in the melting thermograms of the obtained SSPCMs. It is noteworthy that the SSPCMs display a good crystalline performance in the cooling procedure as well. These results indicate the synthesized SSPCMs possess an outstanding phase-change energy-storage behavior, meaning that the phase change behavior of PEG is well-maintained in the molecular structure of the synthesized SSPCMs after

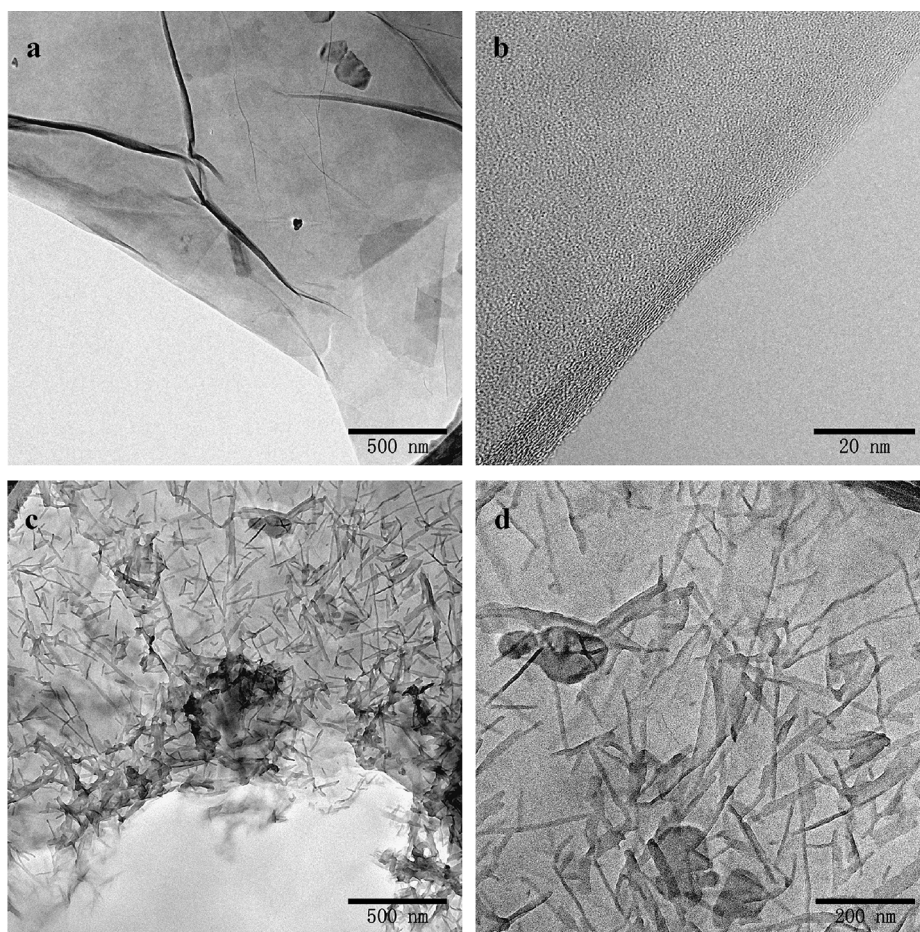


Fig. 3. TEM images of (a, b) the synthesized GO and (c, d) the SSPCM-3.

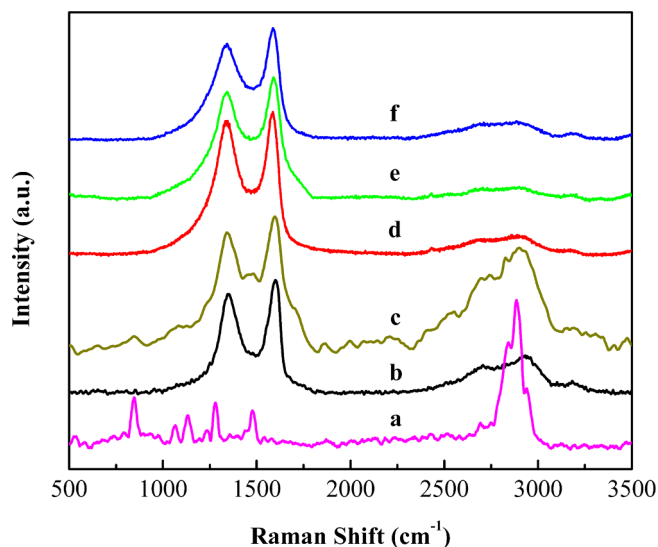


Fig. 4. Raman spectra of (a) pure PEG, (b) GO, (c) SSPCM-1, (d) SSPCM-2, (e) SSPCM-3 and (f) SSPCM-4.

grafted polymerization, providing the strong thermal energy storage properties of the novel SSPCMs. Furthermore, it is evident that the melting temperatures of the samples are lower than that of the pure PEG, and decrease with the increase of GO content in the samples, meaning that the high thermal conductivity of GO enhances the thermal response rate of the SSPCMs. However, as the GO content increases to 12 wt%, the melting temperature goes up to 67.1 °C. This suggests that

the excessive content of GO results in an accumulation in the SSPCMs and breaks the thermally conductive network of the SSPCMs. These results confirm that 9 wt% GO is the optimal additive amount to synthesize the novel SSPCMs, allowing the GO to be dispersed homogeneously in the molecular structure of the novel SSPCMs through self-assembly among GO, PEG, and MDI. The reaction between GO and MDI forms the hard segment of the molecular chain that keeps the solid state of the SSPCMs and has almost no effect on the phase change behavior of PEG. To summarize, the obtained SSPCMs have excellent phase change behaviors, with sample 3 possessing the optimal thermal energy storage properties. It is noteworthy that all the crystallization temperatures of the SSPCMs decrease compared with those of pure PEG, meaning the PEG chains in SSPCMs cannot crystallize perfectly due to the steric hindrance effects of the cross-linking network structure [25,26].

The latent heats of phase transition for all the SSPCMs are slightly lower than those of pure PEG, also confirming that the phase change behavior of PEG is well-maintained in the synthesized SSPCMs after cross-linking polymerization. Thus, the novel SSPCMs possess better thermal energy storage capacities compared with pure PEG. In the three-dimensional network of the composite SSPCMs (Fig. 1), PEG molecular chains are confined by the lamellar structure of the GO skeleton, consequently suppressing and restricting the arrangement and orientation of PEG molecules, partially by steric effects [10,12,25]. Fortunately, this effect does not hinder the thermal energy storage properties of the SSPCMs, yet it keeps the solid state of the PCMs during the phase change procedure, indicative of a novel kind of SSPCM. From Table 1, there is little difference in the heats of phase transition among the synthesized samples, with SSPCM-3 possessing the biggest latent heat during the melting and cooling processes. This further implies that 9 wt% GO dispersed homogeneously in the molecular structure of the

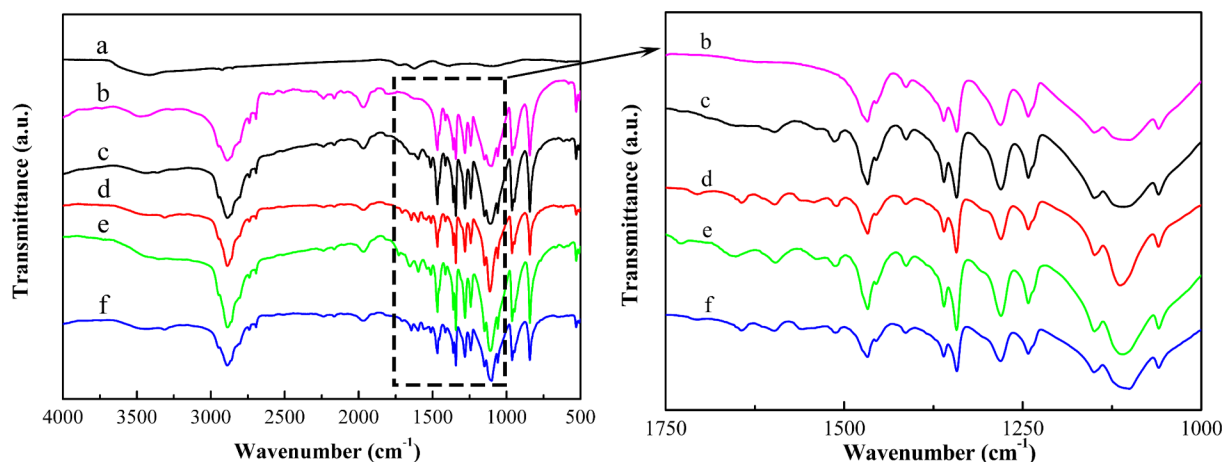


Fig. 5. FTIR spectra of (a) GO, (b) PEG, (c) SSPCM-1, (d) SSPCM-2, (e) SSPCM-3 and (f) SSPCM-4.

novel SSPCMs forms a connected thermally conductive network, and provides the excellent thermal energy storage performance of the novel SSPCMs. These results also confirm that 9 wt% GO is the optimal additive amount to synthesize the SSPCMs.

In addition, a comparison was made of the latent heat of pure PEG, the present SSPCMs and other SSPCMs reported in the literature to highlight the enhancement of the thermal performance of the obtained SSPCMs based on the novel lamellar structure induced by GO, as shown in Table 2. It is worth noting that the experimental heat of melting and

heat of melting ratios (namely the ratio between the heat of melting of the SSPCM and pure PEG) [12,13] at a DSC scanning rate of 5 °C/min in this work are much higher than those of most recently reported PEG-based polymeric SSPCMs. These improvements are mainly ascribed to the introduction of highly thermal conductive GO in the molecular chain of the novel SSPCMs creating a regular lamellar structure established through self-assembly between GO nanosheets and polymer chains in the SSPCMs, leading to an effective thermally conductive network for the interior PEG molecules. As a result, PEG can quickly

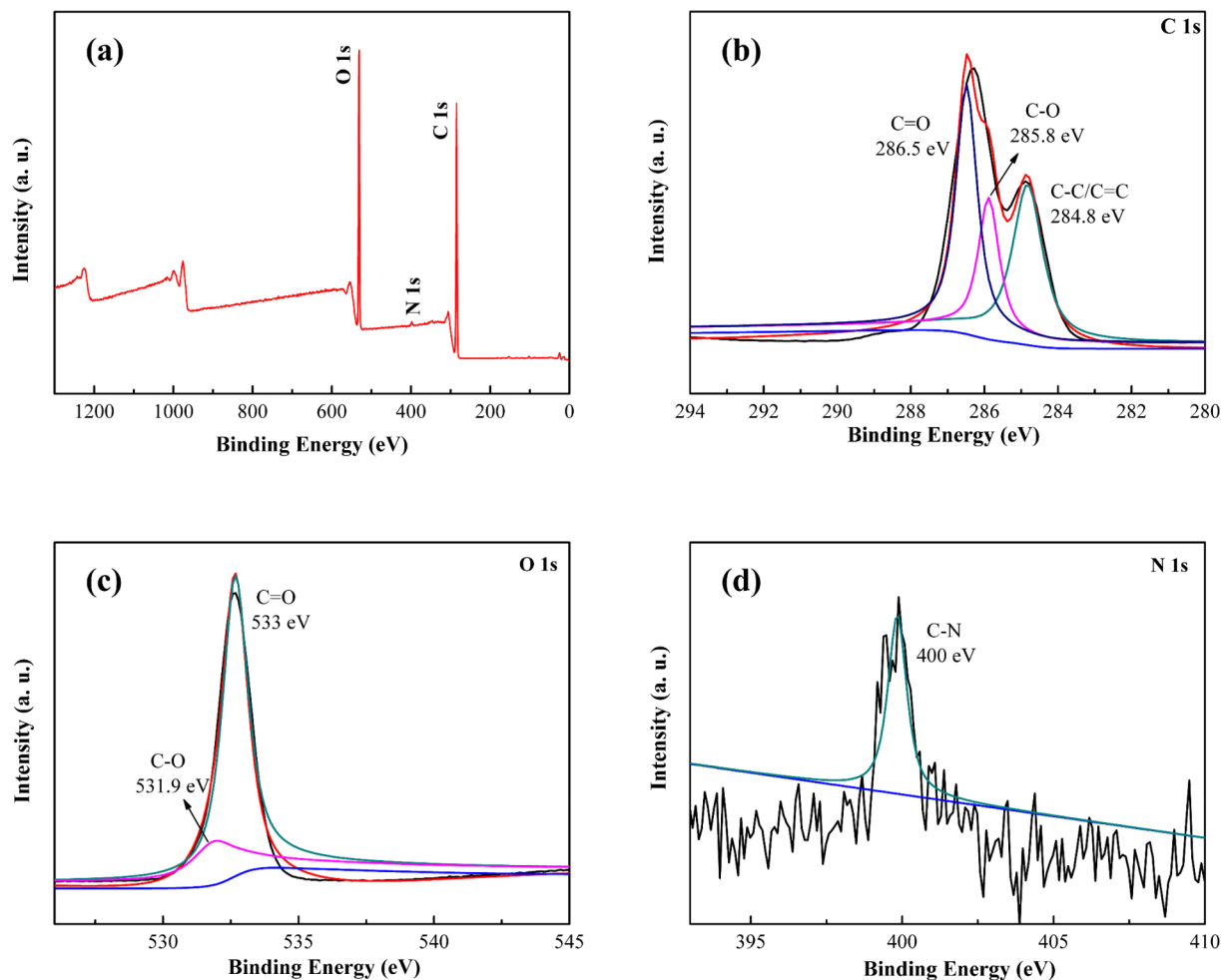


Fig. 6. (a) XPS survey spectrum of SSPCM-3 and high-resolution spectra of SSPCM-3 (b) C 1s, (c) N 1s, (d) O 1s.

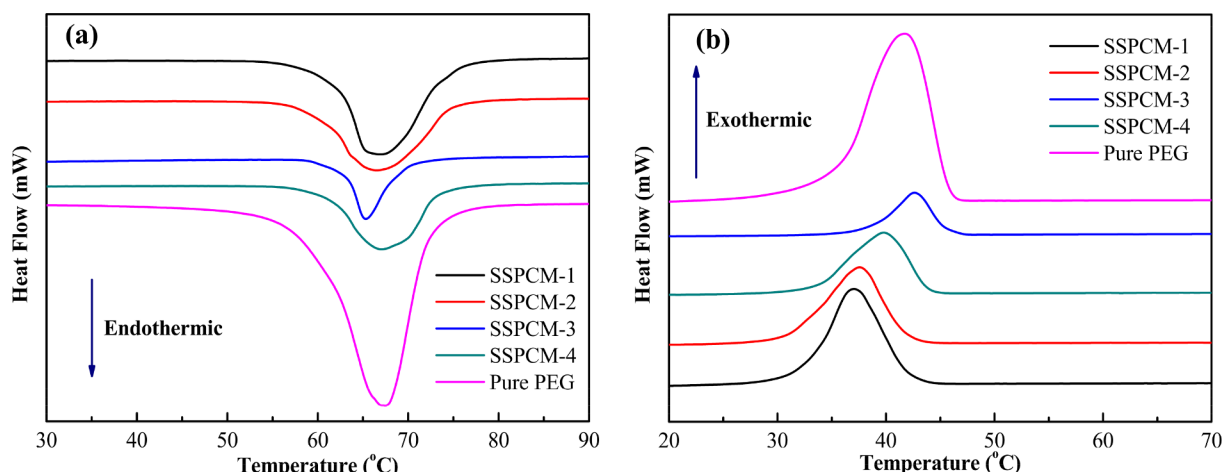


Fig. 7. DSC thermograms of the obtained SSPCMs and pure PEG.

Table 1
Phase change behavior of the synthesized SSPCMs.

Sample name	Melting		Crystallization	
	Temperature (°C)	Heat of phase transition (J/g)	Temperature (°C)	Heat of phase transition (J/g)
SSPCM-1	66.9	146.1	37.0	146.1
SSPCM-2	66.5	147.3	37.6	147.0
SSPCM-3	65.3	158.2	42.6	157.5
SSPCM-4	67.1	136.0	39.8	134.3
Pure PEG	67.5	170.7	41.8	167.1

respond to temperature variation and fully change its phase due to having adequate space for its segment mobility supplied by the lamellar structure of the SSPCMs, enhancing the material's properties.

Based on previous reports [37–40], to further affirm the phase change behavior of the SSPCMs, the degree of crystallinity (F_c) of the prepared samples was calculated by the following Eq. (1),

$$F_c = \frac{\Delta H_{m, comp}}{\Delta H_{m, PCM}^0 \beta} \times 100\% \quad (1)$$

where $\Delta H_{m, comp}$ is the heat of melting of the SSPCMs, $\Delta H_{m, PCM}^0$ is the heat of fusion of 100% crystalline PEG (196.8 J/g [41,42]), and β is the content of soft segments, calculated using the feed weight. The F_c values of PEG in SSPCMs were 81.11% (SSPCM-1), 84.02% (SSPCM-2), 92.65% (SSPCM-3) and 81.72% (SSPCM-4), which is higher than the reported literature [28–36]. These results are attributed to the high

thermal conductivity and regular, lamellar structure of the SSPCMs, induced by GO, that makes the interior PEG molecules crystallize well.

3.5. Thermal cycling stability of the SSPCMs

From the above results, it was decided that SSPCM-3 possesses the best phase change behavior among the samples. Therefore, the thermal cycling test was conducted on this sample to evaluate the thermal cycling stability of the novel SSPCMs for 100 thermal cycles. The obtained DSC curves of the 1st time, 20th time, 50th time, and 100th time are shown in Fig. 8, with the corresponding values derived from the DSC curves listed in Table 3. It can be found that the melting temperature range of the sample had almost no change after 100 thermal cycles. Although the crystalline peak shifts slightly to lower temperatures after the thermal cycles, the main crystalline temperature range still accumulates around 40 °C. Furthermore, it is noteworthy that the latent heat value of the SSPCM-3 just has only a slight change after 100 thermal cycles. This implies that the as-synthesized SSPCM-3 has a good thermal reliability after 100 thermal cycles. This phenomenon is also attributed to the good thermal conductivity performance of the GO-enhanced molecular skeleton that creates a good thermal stability. As a result, the synthesized SSPCMs have a good potential for practical applications in thermal energy storage.

3.6. Thermal regulation of the SSPCMs

To evaluate the improvement in the thermal-regulating behavior of the SSPCMs, temperature-time curves of pure PEG and SSPCM-3 were

Table 2
A comparison of heat of melting among bulk PEG, the present SSPCMs and other PEG-based SSPCMs.

SSPCMs Sample	Heating rate (°C/min)	Heat of melting -PCM (J/g)	Heat of melting -SSPCM (J/g)	Heat of melting ratio (R, %)	Ref.
Melamine/formaldehyde/PEG 6000	10	172.5	109.4	63.42	[10]
β-Cyclodextrin/MDI/PEG 8000	2	148.12	115.20	77.77	[12]
Diethylenetriamine/MDI/PEG 8000	2	148.12	119.90	80.95	[13]
Xylitol/MDI/PEG 6000 (91.44 wt%)	10	178.1	76.37	42.88	[27]
Halloysite nanotubes/HDIB/PEG 6000 (78.4 wt%)	10	182.7	118.7	64.97	[28]
GO/HDIB/PEG 4000	10	182.7	78.0	42.69	[29]
Phloroglucinol/IPDI/PEG 8000	5	Unknown	146.6	Unknown	[30]
Glycerol/MDI/PEG 6000	10	234.4	128.5	54.82	[31]
Poly(styrene-co-maleic anhydride)-graft-PEG 6000	10	221.31	155.21	70.13	[32]
Castor oil/MDI/PEG 6000 (83.5 wt%)	10	196.30	93.23	47.49	[33]
Sorbitol/MDI/PEG 8000 (93.8 wt%)	2	148.1	107.5	72.59	[34]
Glucose/MDI/PEG 8000 (70 wt%)	2	148.12	131.90	89.05	[35]
Tween 80/MDI/PEG 6000 (86.49 wt%)	10	234.4	127.7	54.48	[36]
SSPCM-3	5	170.7	158.2	92.70	This work

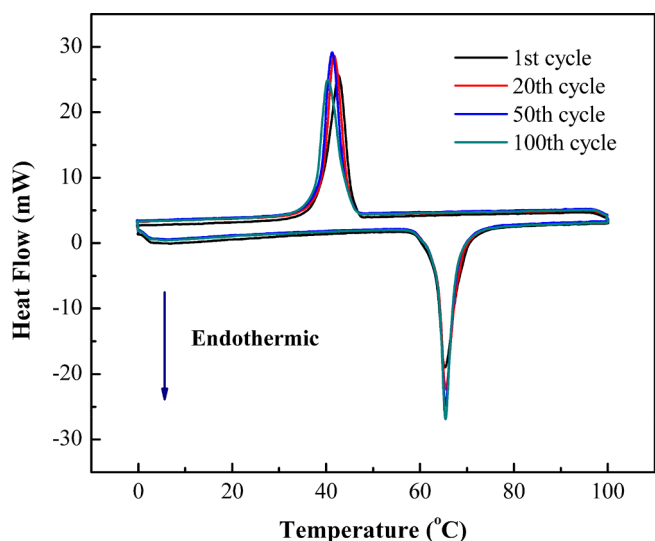


Fig. 8. DSC thermograms for the thermal cycling stability of the SSPCM-3 over 100 thermal cycles.

Table 3
Thermal properties of the SSPCM-3 under thermal cycling.

Cycling number	Melting		Crystallization	
	Temperature (°C)	Heat of phase transition (J/g)	Temperature (°C)	Heat of phase transition (J/g)
1	65.3	158.2	42.6	157.5
20	65.5	158.0	41.7	156.4
50	65.4	157.9	41.3	155.5
100	65.4	157.1	40.4	155.2

investigated in a temperature range from -10°C to 100°C as seen in Fig. 9. It is evident that the temperature of both pure PEG and SSPCM-3 takes as long as 1960 s to reach a thermal balance of 96°C during the heating time. This indicates that the interior PEG of the SSPCMs can be fully molten and absorb the exterior heat through phase change procedures to maintain their temperature. The temperature increasing rates of the SSPCMs are similar than that of pure PEG. Furthermore, when the samples reach their melting points, an inflection point appears in their temperature change curve, indicating that a phase change process took place. It can be distinguished that the temperature of PEG and SSPCM-3 tend to be constant in the temperature range of 50°C

through 57°C , which is near their phase change temperature range. The procedure lasts about 310 s for PEG, while it lasts about 410 s for SSPCM-3. The appearance of this elongated temperature plateau further confirms that the composite PCMs have better thermal regulation properties than PEG due to the thermal conductive enhancement of GO. It also implies that the regular lamellar structure of GO in the polymer's molecular chain acts an effective thermally conductive network for interior PEG molecules by providing enough space for adequate mobility while also preventing PEG leakage. Consequently, the SSPCMs possess an excellent phase change behavior and a great capacity for intelligently controlling thermal regulation. The same results are also present in the cooling temperature-time curves as illustrated in Fig. 9b. The temperature of the pure PEG and SSPCM-3 can be maintained in the range between 45°C and 48°C for approximately 350 s for the PEG and 340 s for SSPCM-3. These results demonstrate that the SSPCMs have a potential application in thermal management for both electronic devices and thermal energy storage systems.

3.7. Crystalline properties of SSPCMs

To reveal the crystalline morphology of the SSPCMs, a powder X-ray diffraction technique (PXRD) was employed to investigate their phase change behavior at room temperature, with the obtained patterns of the synthesized GO, pure PEG, and SSPCM-3 shown in Fig. 10. It can be seen that the characteristic diffraction peak of GO occurs at around 11° , corresponding to the typical sharp peak of (002) due to the intercalation of functional groups into the graphite layers. This result indicates that GO has been successfully prepared. PEG displays two sharp intense peaks at 19.2° and 23.3° , which are attributed to crystal reflection planes (1 2 0) and concerted (1 1 2), (0 3 2) planes, respectively, confirming its highly crystalline structure [43–45]. The diffraction patterns of the SSPCM-3 are similar to those of pure PEG, which demonstrates that pristine PEG and SSPCMs have similar crystalline structure and crystal cell types. These results imply that PEG molecules maintain a good crystalline structure through the intercalation of functional groups in the lamellar structure of the GO, which enlarges space for PEG mobility and decreases the confinement of the hard segment of the molecular chain in the SSPCMs.

In-situ, temperature-dependent XRD measurements were also performed on the representative SSPCM-3 samples to further define the crystalline properties of the synthesized SSPCMs. The obtained diffraction patterns were recorded at different temperatures from 25 to 90°C , seen in Fig. 11. During the heating progress, it can be clearly seen that the SSPCM-3 shows similar diffraction patterns from 25 to 55°C when compared with the PXRD patterns at room temperature, with the position of the two main diffraction peaks remaining unchanged. The

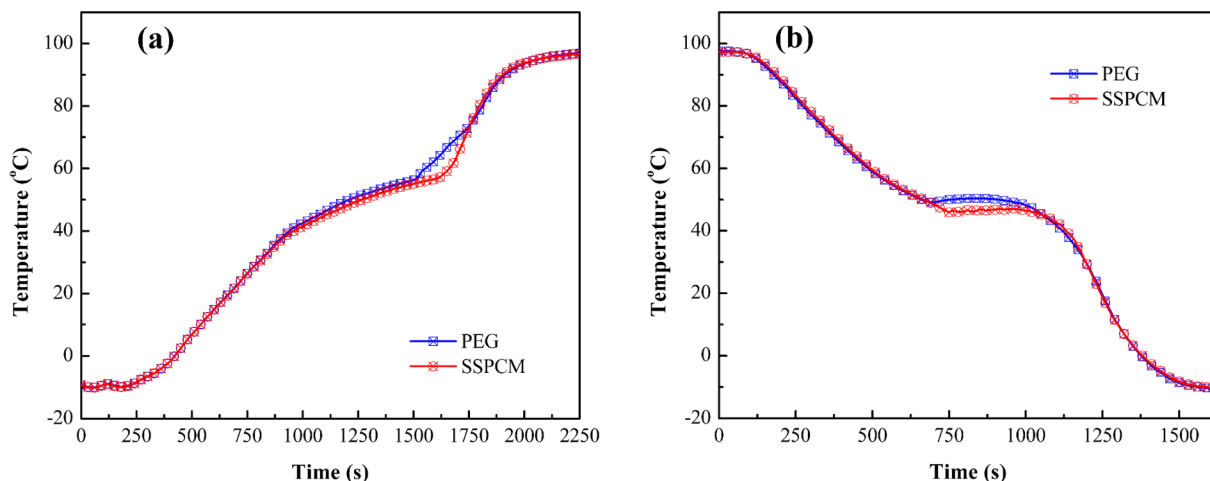


Fig. 9. Temperature-time curves of pure PEG and SSPCM-3: (a) heating curves, (b) freezing curves.

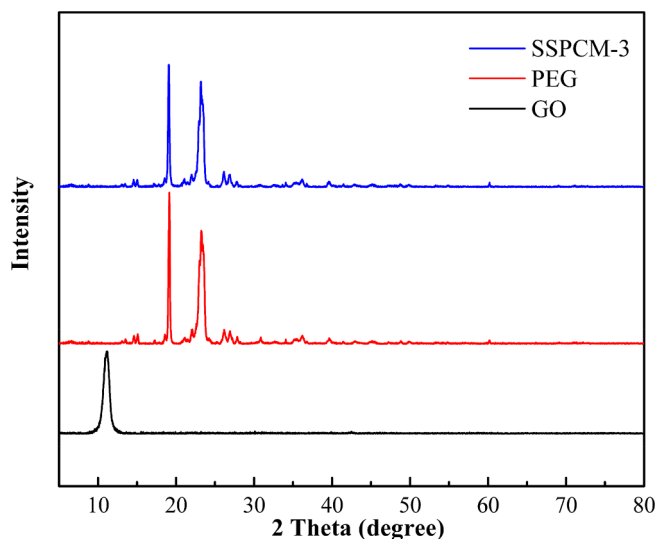


Fig. 10. XRD patterns of GO, pure PEG and SSPCM-3.

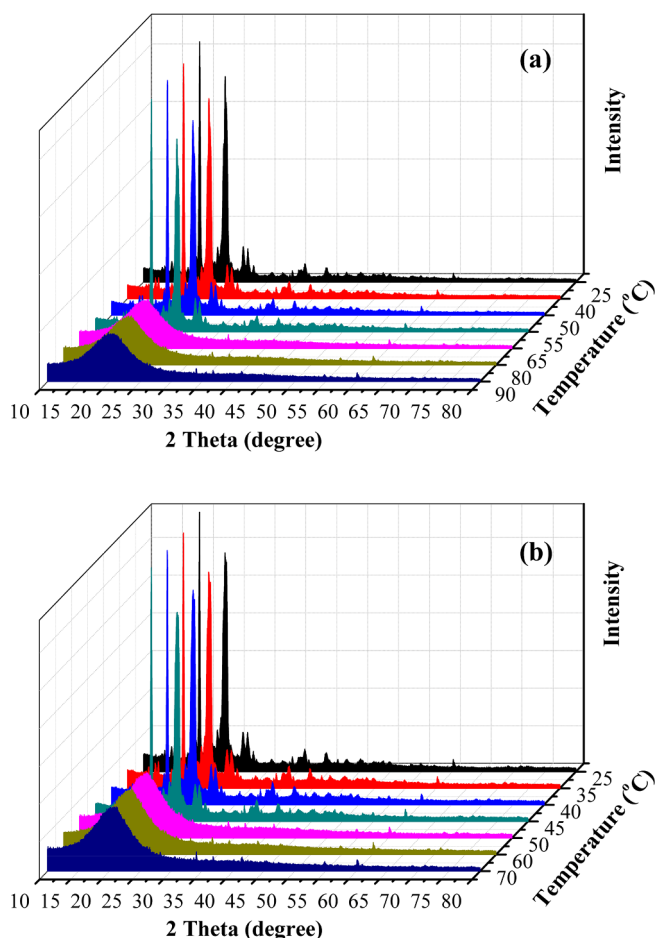


Fig. 11. In-situ temperature-dependent XRD profiles of SSPCM-3: (a) heating progress, (b) cooling process.

XRD profiles after 65 °C reflects a fully amorphous state for the SSPCM-3. This suggests that the SSPCMs display a good crystalline structure within 25–55 °C and completely melt beyond 65 °C. Furthermore, these crystalline diffraction peaks for SSPCM-3 occur again as the temperature was decreased to 45 °C during the cooling process as shown in Fig. 11(b), with the most prominent reflections being quite similar to the diffraction patterns of pure PEG. These results indicate that the

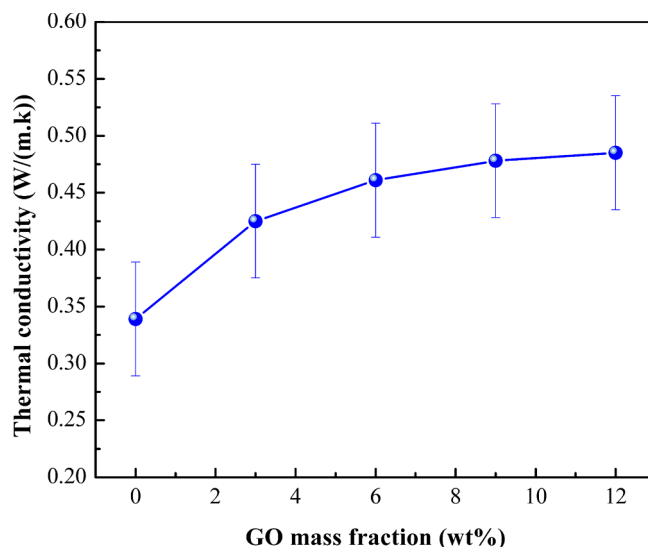


Fig. 12. Thermal conductivity of the obtained SSPCMs and pure PEG.

SSPCMs possess good phase change behavior with their phase change temperature in the range of 25–55 °C. These results are quite consistent with those obtained from DSC curves. As a result, these findings indicate that the as-synthesized SSPCMs possess an excellent phase change behavior and a good thermal cycling stability.

3.8. Thermal conductivity of the obtained SSPCMs

The thermal conductivity of composite PCMs is considered to be a critical indicator to examine their thermal-energy storing and releasing rates. Therefore, the thermal conductivities of pure PEG and the SSPCMs are shown in Fig. 12. It is evident that the SSPCMs have a much higher thermal conductivity than pure PEG. This result confirms that the GO sheets, which make up the key skeletal material in the hard segment of the SSPCMs molecular chain, effectively enhance the thermal conductivity of the novel SSPCMs. As presented above, the GO sheets and the PEG-MDI prepolymer formed a three-dimensional interconnected network. These highly thermally conductive GO sheets supply a continuous path for thermal energy to transfer throughout the SSPCMs, rapidly. Consequently, the thermal conductivity jumps from 0.338 W·m⁻¹·K⁻¹ (pure PEG) to 0.478 W·m⁻¹·K⁻¹ (SSPCM-3), a 41.4% improvement using 9 wt% GO, which is consistent with previous works [8,24]. In a word, the addition of GO effectively improves the thermal conductivity of the SSPCMs, ensuring that the synthesized SSPCMs have fast thermal responsive rates, and keep excellent phase change energy storage characteristics and good thermal cycling stability.

3.9. Temperature response behavior of the obtained SSPCMs

The temperature response behaviors of the pure PEG and SSPCM-3 samples were further examined using an infrared camera to evaluate their thermal response rates and thermal transfer capacities, as shown in Fig. 13. During heating process, it can be clearly seen that the color of SSPCM-3 changed much faster than that of pure PEG from the temperature distribution images at 0, 20, 40, 60, 80 and 100 s, indicating that the thermal response rate of the SSPCMs is much faster due to their high thermal conductivity enhanced by GO. Therefore, the SSPCMs can respond quickly to changes in the temperature of the external environment and exhibit a good, intelligent temperature-control effect. As expected, the surface temperature of SSPCM-3 gradually declined at a faster rate and remained lower than that of pure PEG during the cooling procedure, based on temperature distribution images. The high thermal conductivity of SSPCM-3 allows heat to transfer rapidly during heating

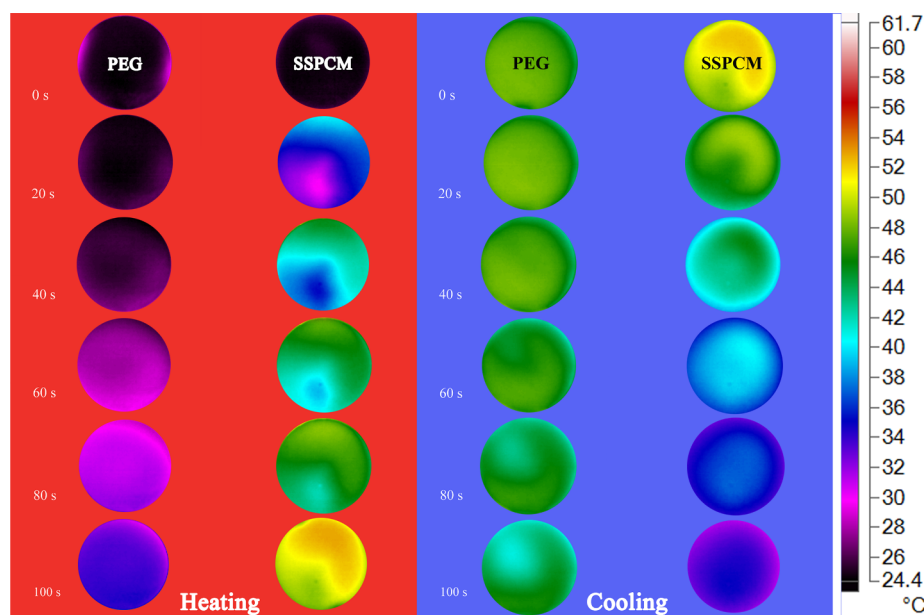


Fig. 13. Thermal images of pure PEG (left) and SSPCM-3 (right) during heating and cooling.

and cooling. The above results also imply that introduction of highly thermally conductive GO in the molecular chains of SSPCMs, resulting in a remarkably fast thermal response rate for them. All these results illustrate that the novel SSPCMs have a great potential application for thermal energy storage systems.

4. Conclusions

In this study, a series of novel polymeric SSPCMs were successfully synthesized using PEG as the phase change ingredient, MDI as the cross-linking agent, and GO as the skeleton material. TEM and SEM images reveal that the novel SSPCMs possess a regular, lamellar structure consisting GO nanosheets with PEG homogeneously intercalated into the sheets through the self-assembly of functional groups. DSC curves indicate that the synthesized SSPCMs possess high latent heat of phase transition that are almost equal to those of pure PEG. Furthermore, the phase-change temperatures and latent heats of the SSPCMs have little change after 100 thermal cycles, displaying a good thermal cycling stability. Thermal regulating experiments showed that the SSPCMs possess a good intelligent temperature-control performance and can keep the temperature in the range of 50 °C–57 °C for about 410 s during a heating process. Moreover, both PXRD and *in-situ* XRD experiments further confirm that the SSPCMs maintain the same crystalline structure as pure PEG, giving the SSPCMs good phase change behaviors. Thermal conductivity experiments confirmed that the highly thermally conductive GO sheets effectively enhance the thermal conductivity of the SSPCMs and supply them with a fast thermal-responsive rate. In summary, the prepared SSPCMs with high phase change latent heats have great potential applications in thermal energy storage.

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